

Embodied Laugh Track: Supporting VR Comedy Viewing Experience with Multimodal Laughter from Virtual Audience

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Figure 1: Embodied Laugh Track system's virtual environment. Users watch streams in a virtual theater through an HMD, where audiences express multimodal laughter (facial expressions, body movements, and sounds) at humorous moments.

Abstract

Virtual reality (VR) comedy viewing with virtual audiences offers promising opportunities for social entertainment experiences. However, implementing convincing social interactions with virtual audiences remains challenging, as limited social cues diminish social presence and the viewing experience. We present “Embodied Laugh Track,” a VR system in which users are immersed in a virtual theater with virtual audiences. The system's virtual audiences express laughter through facial expressions, body movements, and vocal sounds in response to detected humorous moments in the content. In a controlled experiment (N=18), we found that multimodal cues from virtual audiences enhanced users' social presence. While no significant differences were found in engagement and enjoyment,

follow-up interviews revealed that virtual audiences were most effective at amplifying the experience of laughing when synchronized with the moments when users experienced humor. This work contributes to understanding shared viewing experiences with virtual audiences in VR and emotion contagion through multimodal avatar expressions.

CCS Concepts

• Human-centered computing → Virtual reality.

Keywords

Laugh Track, Co-viewing Experiences, Social Presence, Emotional Contagion, Virtual Reality

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1 INTRODUCTION

Online streaming platforms have transformed how we consume entertainment content, enabling new social viewing experiences through remote audience interactions. Viewers often influence each other's emotional responses through real-time comments and reactions, creating a sense of shared experience despite physical separation [18, 19]. Virtual reality (VR) technologies further expand these possibilities by enabling viewers to share virtual environments as embodied avatars, an area of growing research interest [7, 26, 33, 39]. These social viewing experiences hold particular importance for comedy content, where audience reactions play a crucial role in enhancing the viewing experience, as studies have shown that watching comedy with others elicits more laughter than watching alone [4, 45].

However, creating engaging social experiences in virtual environments remains challenging due to limited social cues between remote viewers. The effectiveness of social viewing experiences relies on the transmission of social cues among viewers. While television shows have historically used laugh tracks to enhance viewers' experience [15], these systems rely solely on auditory cues, ignoring the diverse nonverbal cues accompanying laughter. Research shows that human laughter naturally combines vocal sounds with facial expressions and body movements [6], and these visual and auditory cues can independently trigger laughter through its contagious nature [31]. Studies have demonstrated that combining visual and auditory laughter cues produces more potent effects than either modality alone [8].

These findings suggest the potential of incorporating multimodal laugh tracks in virtual environments to enhance comedy viewing experiences. To explore the potential of multimodal laugh tracks, we present "Embodied Laugh Track," a VR system that enhances comedy viewing experiences by surrounding the user with virtual audiences. In this system, users wearing a head-mounted display (HMD) are immersed in a virtual theater where they can watch YouTube Live streams with virtual audiences. When humorous moments are detected in YouTube Live comments, virtual audiences express their laughter not only via vocal sounds but also via facial expressions and body movements. Through these multimodal expressions of virtual audiences, our system aims to create a high sense of social presence as if watching comedy with other audiences, enhancing user engagement and enjoyment.

We conducted a within-participants experiment ($N=18$) comparing six conditions, combining three levels of visual cues (hidden, idle, expressive) and two levels of auditory cues (absent, present) to evaluate the system. Results revealed that perceived social presence increased with higher levels of visual cues and the presence of auditory cues.

While no significant differences were found in engagement and enjoyment, follow-up interviews revealed that temporal alignment between participant and virtual audience reactions was important. Participants reported increased engagement and enjoyment when virtual audiences laughed in synchronization with their reactions but reported diminished viewing experience when the timing of laughter was misaligned. This work contributes to facilitating shared viewing experiences with virtual audiences and emotion contagion through multimodal expressions of avatars in VR.

2 RELATED WORK

2.1 Shared Viewing Experience with Virtual Audiences

The presence of co-viewers enhances emotional engagement and viewing experience during content consumption [20, 40]. To leverage this social effect, various VR systems have been developed to enable shared viewing experiences with social companions remotely. Early attempts at VR co-viewing focused on the photorealistic representation of viewers to create the impression of sitting next to each co-viewer [7, 26]. Later research explored avatar-mediated co-viewing [33], an approach now adopted by commercial platforms like BigscreenVR¹ that offer shared viewing experiences in virtual theaters. Notably, studies have demonstrated that even computer-controlled avatars can enhance the quality of the viewing experience through simulated co-presence [39]. Recent work in live concert viewing has shown that virtual audiences mirroring user movements and generating beat-synchronized behaviors can create engaging experiences [43]. While these studies demonstrate the effectiveness in enhancing presence, the emotional impact of virtual audiences on viewers remains understudied.

2.2 Laughter Contagion

Laughter is highly contagious, as simply seeing others laugh can trigger laughter in those observing [31, 37]. Furthermore, when consuming humorous stimuli with others who laugh, people tend to laugh more compared to when they are alone [4, 45]. These social effects indicate that laughter functions as a signal to elicit positive emotions [14, 31]. Recognizing its importance, the ILHAIRE Project aimed to detect and analyze human laughter [10, 25]. Additionally, research has explored design guidelines for avatar agents' laughter, considering gelotophobia, a trait that refers to the fear of being laughed at [36].

Recent research has explored interfaces that leverage emotion contagion to enhance people's emotional states. Visual approaches include designing mirrors to transform self-reflection, thus inducing positive emotions [1, 44]. Furthermore, video conferencing systems were also designed to enhance facial expressions for increasing social connectedness between remote interlocutors [38]. For auditory approaches, laugh tracks remain a well-established method, with recent systems detecting users' initial laughter and trigger playback to induce positive emotion [13, 15]. While these applications demonstrate the effectiveness of visual and auditory cues independently, research indicates that laughter is multimodal [6] and becomes more contagious when multiple modalities are combined [8]. This motivated us to design a multimodal laugh track system for an avatar to enhance the enjoyment while viewing comedy in VR.

3 SYSTEM DESIGN

3.1 Cues of Virtual Audience Laughter

Laughter comprises three primary features: (A) facial expressions, (B) body movements, and (C) vocal sounds [6]. In our design, we incorporated visual (A, B) and auditory (C) cues into the avatar's interface.

¹Big Screen VR <https://www.bigscreenvr.com/software/>



(a) Hidden Audience (HA): avatars are invisible to the user, showing only the virtual theater environment.

(b) Idle Audience (IA): avatars are visible and perform idle animations to indicate presence.

(c) Expressive Audience (EA): avatars express laughter through both facial expressions and body movements.

Figure 2: Three levels of visual cues for virtual audience feedback. These visual conditions were combined with two levels of auditory cues (absent/present), resulting in six experimental conditions.

3.1.1 Visual Cues. The facial expressions of an avatar were implemented using the Facial Action Coding System (FACS), a comprehensive framework that quantifies human facial movements through Action Units (AUs) [11]. AUs were defined as the contractions and relaxations of facial muscles. Based on FACS research on laughter expressions, we implemented mouth opening (AU25) and jaw lowering (AU26) movements [3, 9]. Furthermore, natural laughter, known as the Duchenne smile, involves movement of the orbicularis oculi muscle [28, 35]. Accordingly, we implemented upper lid raising (AU5), cheek raising (AU6), lid tightening (AU7), and nose wrinkling (AU9) [34]. We used avatars from Character Creator 4², also used to control their facial expressions based on AUs.

For the body movements of an avatar, we adopted the “Sitting clapping” animation distributed by Mixamo³. Research has shown that characteristic laughter movements primarily consist of shoulder rotation, hand movements, and head movements, with these patterns remaining consistent across both standing and sitting postures [6, 16, 17]. We ensured that the selected animation reproduced these characteristics. The animation duration was approximately 4 seconds, so facial expression animations were synchronized to match this length. The avatar implementation, featuring facial expressions and body movements, was shown in Fig. 2 (c).

3.1.2 Auditory Cues. Avatar laughter sounds were from a dataset containing 7,489 laughter samples collected from online videos [42]. To match the animation duration, we extracted twenty 4-second single-speaker laughter samples, comprising ten samples each from

male and female speakers. Additionally, to reproduce acoustic spatial characteristics, we implemented volume attenuation based on the distance between the user and laughing avatars:

$$V = V_0 \cdot \left(1 - \frac{d - d_{min}}{d_{max} - d_{min}}\right) \quad (1)$$

where V_0 is initial volume, d is source distance, with empirically chosen bounds $d_{min} = 1$ m and $d_{max} = 20$ m.

3.2 Virtual Theater Environment

We modeled our virtual theater after Yoshimoto ∞ Dome⁴ STAGE, one of the local popular comedy theaters of the authors’ country. The theater space is configured as shown in Fig. 1, with 30 virtual audience members seated across three rows. The user’s avatar was positioned in the center of the third row to optimize the visibility of other audience members’ facial expressions and movements. The avatar’s hips and feet of the users were fixed to the seat, with posture controlled through inverse kinematics based on head and hand tracking data.

The front-facing screen supports live YouTube Live streams’ playback and archived content. When the system detected the associated live comments in the video that indicated laughter (“w”, “笑”, “草”, common text expressions equivalent to “LOL”), it randomly selected one virtual audience member per comment to trigger a laughing animation. The system was implemented in Unity and operates as a standalone application on VR head-mounted displays.

²Character Creator 4 <https://www.reallusion.com/jp/character-creator/>

³Mixamo <https://www.mixamo.com/>

⁴Yoshimoto ∞ Dome <https://mugendai-dome.yoshimoto.co.jp/about>

4 EVALUATION

4.1 Experiment Design

We conducted a within-participant experiment to evaluate how different amounts of visual and auditory cues of virtual audience laughter affected users' comedy viewing experience in VR. Building on prior work on multimodal aspects of contagious laughter [8], we examined two independent variables: *visual* and *auditory* cues of virtual audience laughter. The first variable, visual cue, consisted of three levels (Fig. 2). (a) Hidden Audience (**HA**): All avatars were invisible, showing only the theater environment, (b) Idle Audience (**IA**): Avatars were visible and performed idle animations to maintain the social presence and (c) Expressive Audience (**EA**): Avatars expressed laughter through synchronized facial expressions and body movements. The second variable, auditory cue, consisted of two levels: absence (**No Auditory Cues**) or presence (**Auditory Cues**) of laughter sounds. This led to a 3×2 factorial design. Each participant experienced all six conditions in a counterbalanced order using a 6×6 Latin square design to minimize order effects.

4.2 Video Stimuli

As video stimuli, we selected six comedy clips from the archives of JaruJaru Tower⁵, one of the most popular YouTube Live comedy channels in the country where the study was conducted. Each clip was edited to 2 minutes and 30 seconds to maintain consistent length. To ensure a fair comparison across conditions, we first analyzed the number of laughter comments in each selected clip and found that the clip with the fewest had 57 instances. We then standardized all clips to trigger virtual audience laughter 57 times to maintain consistent frequency across conditions.

4.3 Participants

We recruited 18 participants in this study, consisting of 13 males and 5 females, with an average age of 32.2 years ($SD = 9.1$ years) through social networking services. All participants had normal or corrected-to-normal vision and normal hearing. To prevent response bias, we initially told participants that the study aimed to "analyze user behavior when watching videos in virtual environments." We debriefed participants about the actual purpose of the experiment after they had experienced all the trials. The participants were paid \$10 after completing the 60-minute experiment. The experimental protocol was approved by the local ethics committee.

4.4 Procedure

Participants were briefed on the experiment overview and they signed a consent form before the experiment started. Before the VR experience, participants completed the Simulator Sickness Questionnaire [32] to record their physical condition. Participants wore an HMD and received instructions about the virtual environment's layout and their avatar control. Participants experienced the experiment using an application that runs on a standalone HMD, allowing them to engage without external devices restricting their movement. As a result, the graphical fidelity of the virtual environment and avatars was relatively simple (as shown in Fig. 1). However,

they were instructed to focus on the comedy clips, minimizing the impact of environmental realism on their experience. Participants had one minute to acclimate to the virtual environment for each of the six conditions. After each comedy clip, participants completed a questionnaire and a one-minute distractor task to minimize carryover effects from the previous condition. The distractor task consisted of a simple working memory task [12], which was unrelated to the comedy viewing but served to reset participants' cognitive state before the next trial. After completing all trials, participants participated in a semi-structured interview, with questions like "How did the presence and reactions of virtual audiences affect your viewing experience?" The entire experimental procedure took approximately 60 minutes.

4.5 Measurements

To evaluate the effectiveness of the multimodal reaction from virtual audiences, we measured participants' social presence, engagement, and enjoyment while watching the comedy in VR. Following prior work on virtual audience co-viewing [43], we adopted the social presence questionnaire from [5]. Engagement and enjoyment were assessed using scales from [23], previously applied to evaluate social viewing experiences. All measures were rated by participants using seven-point Likert scales. We conducted a 3 (visual cues) $\times$ 2 (auditory cues) ANOVA for each measure.

5 RESULTS

5.1 Social Presence

We first performed aligned rank transform (ART) [41], which enabled us to conduct an ANOVA for non-parametric data, and conducted a two-way ANOVA. The result showed a significant main effect of visual cues on social presence ($F(2, 85) = 44.92, p < .001, \eta_p^2 = 0.514$). The main effect of auditory cues on social presence was not significant ($F(1, 85) = 0.63, p = .43, \eta_p^2 = 0.007$). The interaction effect of visual cues and auditory cues on social presence was significant ($F(2, 85) = 39.98, p < .001, \eta_p^2 = 0.49$). Thereafter, we conducted Friedman tests to verify the simple main effects of visual cues⁶, which revealed significant differences between all visual cue levels (**HA** vs. **EA**, **IA** vs. **EA**: $p < .001$; **HA** vs. **IA**: $p < .05$), with higher social presence scores in **EA** condition compared to **LV** and **NV** conditions, suggesting that richer visual feedback enhanced participants' sense of being with others. Moreover, we performed Wilcoxon signed-rank tests to examine the simple main effects of auditory cues, which showed significant differences in all visual cue conditions (**HA**, **IA**: $p < .001$; **EA**: $p < .05$), with an inverse effect observed in **IA** condition.

5.2 Engagement and Enjoyment

After applying ART, we conducted a two-way ANOVA for each measure. For engagement, the result showed no significant main effect of visual cues ($F(2, 85) = 2.23, p = .112$). The main effect of auditory cues was not significant ($F(1, 85) = 0.12, p = .733$). Furthermore, the interaction effect of visual cues and auditory cues was not significant ($F(2, 85) = 0.40, p = .671$). For enjoyment, the

⁵JaruJaru Tower <https://www.youtube.com/@jarujarutower365/>

⁶Tests selected based on levels: Friedman test for visual cues (3 levels) and Wilcoxon signed-rank test for auditory cues (2 levels).

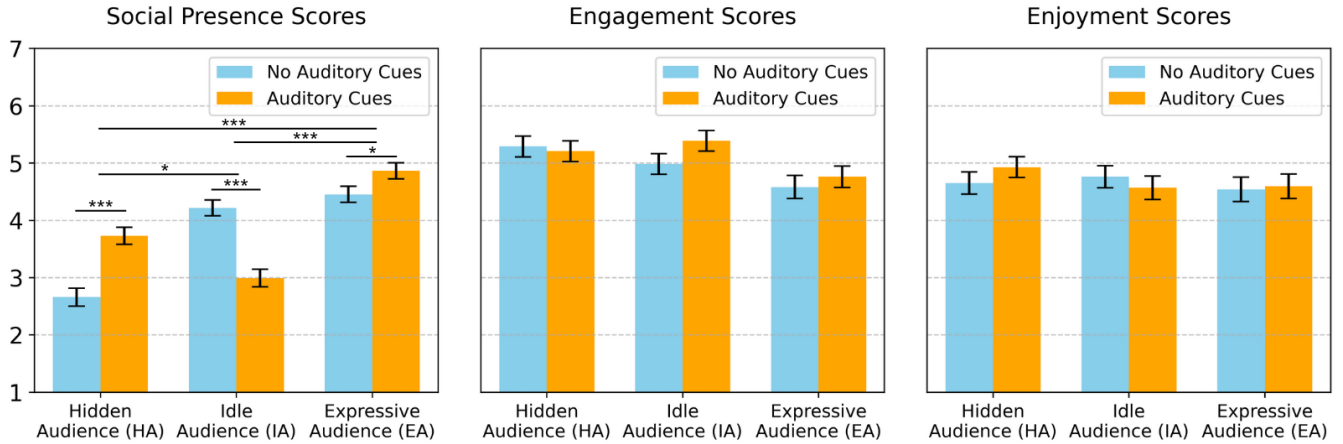


Figure 3: Mean ($\pm$ standard error) scores of Social Presence, Engagement, and Enjoyment across visual (HA: Hidden Audience, IA: Idle Audience, EA: Expressive Audience) and auditory conditions. For Social Presence, we found significant differences between all visual cue levels, and auditory cues enhanced the scores in HA and EA conditions but showed an inverse effect in IA condition. No significant visual or auditory cue effects were found for Engagement and Enjoyment. * $p < .05$, * $p < .001$.**

result showed no significant main effect of visual cues ($F(2, 85) = 0.28$, $p = .757$). The main effect of auditory cues was not significant ($F(1, 85) = 0.07$, $p = .798$). Furthermore, the interaction effect of visual cues and auditory cues was not significant ($F(2, 85) = 0.55$, $p = .580$). Fig. 3 shows significant differences in social presence scores across visual and auditory conditions. However, we found no significant main effects or interactions for either engagement or enjoyment.

5.3 Interview Findings

Participants provided positive feedback about congruent visual-auditory expressions in the EA condition: P09 described: “Having an audience that showed reactions made it feel more like watching in a theater.” P03 reported: “Seeing the audience move was reassuring. I felt like a fellow audience member.” However, regarding the inverse effect of auditory cues in IA condition, participants expressed concerns: P01 noted: “It felt eerily scary, like a paranormal phenomenon.” P18 remarked: “The motionless people didn’t feel real. It was more unsettling than having no one there.”

Several participants reported that misaligned reactions with virtual audiences affected their viewing experience: P12 described: “When virtual audiences laughed at moments I didn’t find funny, it was disorienting.” P07 described: “When virtual audiences laughed at moments I didn’t find humorous, it felt unnatural and distracting.”

6 DISCUSSION

6.1 Effects of Multimodal Expression Cues on Social Presence

Our results demonstrated that participants experienced increased social presence with higher levels of visual and auditory cues from virtual audiences when watching comedy in VR. Notably, the significant difference between HA and EA conditions with auditory feedback suggests the potential to create more social experiences compared to audio-only laugh tracks. Furthermore, our interview

findings indicate that even computer-controlled avatars can enhance social presence when their multimodal expressions are synchronized with the content. Our finding that synchronized multimodal expressions enhance social presence extends previous findings [39, 43] on co-viewing with virtual audiences to emotionally evocative content such as comedy.

However, we observed an interesting inverse effect in IA condition. Specifically, adding auditory cues to static audience avatars resulted in lower social presence than when these cues were absent. This finding aligns with previous research suggesting that inconsistencies between appearance and behavioral realism can diminish social presence in virtual environments [2]. These results provide important implications for the design of virtual audiences in facilitating co-viewing experiences in VR. While a multimodal laugh track system can enhance social presence, maintaining consistency between visual and auditory feedback is equally crucial. This is particularly important because inconsistent feedback can be more detrimental than no feedback. Specifically, when technical constraints limit the implementation of bodily emotional expressions, designers should avoid solutions where static avatars produce only auditory feedback.

6.2 Effects of Audience Reaction Timing on User Experience

Our quantitative analysis showed no significant differences in engagement and enjoyment scores across conditions. Interview findings further provided some clues that virtual audiences’ laughter timing affected the experience in two ways. Participants reported heightened feelings of connection and shared experience when virtual audiences laughed synchronously with them, leading to increased engagement and enjoyment. However, when virtual audience reactions were misaligned with participants’ natural response timing, they reported feeling distracted and disconnected, resulting in decreased engagement.

Several approaches could be implemented to address this misalignment issue. First, improving the accuracy of humorous moment annotation would enhance the synchronization of virtual audience reactions. While our current implementation uses YouTube Live comments as triggers for virtual audience actions, converting text-based comments into appropriate laughter expressions has inherent limitations. Automated detection of humorous moments using machine learning models (e.g., [22, 24, 29]) could significantly enhance experience provided by the system. Second, this system could be adapted for asynchronous co-viewing with close friends, leveraging the natural comfort and increased co-laughter observed between familiar pairs [21]. This adaptation would be particularly valuable for friends and family members who want to share comedy experiences but face scheduling conflicts or time zone differences. By recording the moments when each viewer laughs during their viewing sessions, the system could create an asynchronous shared viewing experience even when friends cannot watch comedy content simultaneously. Third, we could implement real-time audience reaction adjustment based on automatic detection of user laughter through multimodal sensing [13, 27]. This approach would move beyond simple comment-based triggers to create more synchronized interactions by detecting users' laughter through facial expressions, body movements, and vocal sounds.

7 LIMITATION AND FUTURE DIRECTIONS

The main limitation of this study is that the experiments were conducted in a laboratory setting. This setting may have caused participants to suppress their laughter, potentially limiting their natural reactions to comedy content. Additionally, the limited duration of the experiment sessions and the use of pre-selected comedy clips may not fully represent the diverse ways people naturally consume comedy content. While the controlled experiment gave us an initial understanding of the effect of multimodal laugh tracks on the comedy viewing experience in VR, future explorations using field experiments are encouraged.

The widespread adoption of social VR platforms like VRChat demonstrates how VR technology is becoming integrated into users' daily social interactions [30]. Implementing our system on these platforms could enable us to observe user interactions with virtual audiences or remote co-viewers in more natural, daily-use scenarios. Furthermore, the asynchronous viewing application discussed in the previous section shows promise. We see the potential that the design of multimodal laugh tracks for the avatar could facilitate asynchronous content consumption with family members or friends in different time zones. Our future work is to examine the effect of this multimodal laugh track system on the co-viewing experience with remote interlocutors in social VR.

8 CONCLUSION

This study investigated how visual and auditory cues from virtual audiences affect social presence, enjoyment, and engagement in VR comedy viewing experiences. Our results demonstrated that multimodal cues from virtual audiences enhanced social presence, particularly when visual and auditory feedback were synchronized with viewer's laughter moment. While we found no significant

differences in enjoyment and engagement, interview findings revealed the importance of temporal alignment between the reactions from the virtual audience and the viewer. These insights contribute to our understanding of emotion contagion through multimodal virtual agents in creating engaging co-viewing experiences in VR. Our findings suggest promising directions for future social VR applications, particularly in scenarios where spatial and temporal constraints prevent co-located viewing experiences.

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References

- [1] Soyoung Ahn and Yongsoo Choi. 2023. Smiley: Designing Smile Recognition Smart Mirror for Premenstrual Syndrome. *International Journal of Human-Computer Interaction* 39, 12 (2023), 2401–2419. <https://doi.org/10.1080/10447318.2022.2075638>
- [2] Jeremy N. Bailenson, Kim Swinth, Crystal Hoyt, Susan Persky, Alex Dimov, and Jim Blascovich. 2005. The Independent and Interactive Effects of Embodied-Agent Appearance and Behavior on Self-Report, Cognitive, and Behavioral Markers of Copresence in Immersive Virtual Environments. *Presence* 14, 4 (Aug 2005), 379–393. <https://doi.org/10.1162/105474605774785235>
- [3] Ursula Beermann, Fabian Gander, Damian Hildebrand, Tobias Wyss, and Willibald Ruch. 2009. Laughing at oneself: Trait or state?. In *Current and Future Perspectives in Facial Expression Research: Topics and Methodical Questions*, E. Bnninger-Huber and D. Peham (Eds.). Innsbruck University Press, Innsbruck, Austria, 31–36. <https://doi.org/10.5167/uzh-20237>
- [4] Lauren K. Bush, Carol L. Barr, Gregory J. McHugo, and John T. Lanzetta. 1989. The effects of facial control and facial mimicry on subjective reactions to comedy routines. *Motivation and Emotion* 13, 1 (01 Mar 1989), 31–52. <https://doi.org/10.1007/BF00995543>
- [5] Jaclyn Ann Cameron. 2016. *The on-screen water cooler: Effects of televised user-generated comments on cognitive processing, social presence, and viewing experience*. Ph.D. Dissertation. University of Tennessee. https://trace.tennessee.edu/utk_graddiss/3899
- [6] Sarah Cosentino, Salvatore Sessa, and Atsuo Takanishi. 2016. Quantitative Laughter Detection, Measurement, and Classification: A Critical Survey. *IEEE Reviews in Biomedical Engineering* 9 (2016), 148–162. <https://doi.org/10.1109/RBME.2016.2527638>
- [7] Francesca De Simone, Jie Li, Henrique Galvan Debarba, Abdallah El Ali, Simon N.B Gunkel, and Pablo Cesar. 2019. Watching Videos Together in Social Virtual Reality: An Experimental Study on User's QoE. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 890–891. <https://doi.org/10.1109/VR.2019.8798264>
- [8] Micaela De Weck, Benot Perriard, Jean-Marie Annoni, and Juliane Britz. 2022. Hearing Someone Laugh and Seeing Someone Yawn: Modality-Specific Contagion of Laughter and Yawning in the Absence of Others. *Frontiers in Psychology* 13 (2022). <https://doi.org/10.3389/fpsyg.2022.780665>
- [9] Philipp Drack, Tania Huber, and Willibald Ruch. 2009. The apex of happy laughter: A FACS-study with actors. In *Proceedings of the International Meeting at the Institute of Psychology*. Innsbruck University Press, Innsbruck, Austria. <https://doi.org/10.5167/uzh-20236>
- [10] Stéphane Dupont, Hüseyin Çakmak, Will Curran, Thierry Dutoit, Jennifer Hofmann, Gary McKeown, Olivier Pietquin, Tracey Platt, Willibald Ruch, and Jérôme Urbain. 2016. *Laughter Research: A Review of the ILHAIRE Project*. Springer International Publishing, Cham, 147–181. https://doi.org/10.1007/978-3-319-31056-5_9
- [11] Paul Ekman and Wallace V Friesen. 1978. Facial action coding system. *Environmental Psychology & Nonverbal Behavior* (1978). <https://doi.org/10.1037/d27734-000>
- [12] Jeffrey L. Foster, Zach Shipstead, Tyler L. Harrison, Kenny L. Hicks, Thomas S. Redick, and Randall W. Engle. 2015. Shortened complex span tasks can reliably measure working memory capacity. *Memory & Cognition* 43, 2 (01 Feb 2015), 226–236. <https://doi.org/10.3758/s13421-014-0461-7>
- [13] Shogo Fukushima, Yuki Hashimoto, Takashi Nozawa, and Hiroyuki Kajimoto. 2010. Laugh enhancer using laugh track synchronized with the user's laugh motion. In *CHI '10 Extended Abstracts on Human Factors in Computing Systems (Atlanta, Georgia, USA) (CHI EA '10)*. Association for Computing Machinery, New York, NY, USA, 36133618. <https://doi.org/10.1145/1753846.1754027>
- [14] Matthew Gervais and David Sloan Wilson. 2005. The Evolution and Functions of Laughter and Humor: A Synthetic Approach. *The Quarterly Review of Biology* 80,

- 4 (2005), 395–430. <https://doi.org/10.1086/498281>
- [15] Gina Giotta. 2017. Sounding Live: An Institutional History of the Television Laugh Track. *Journal of Communication Inquiry* 41, 4 (2017), 331–348. <https://doi.org/10.1177/0196859917713760>
- [16] Harry J. Griffin, Min S.H. Aung, Bernardino Romera-Paredes, Ciaran McLoughlin, Gary McKeown, William Curran, and Nadia Bianchi-Berthouze. 2013. Laughter Type Recognition from Whole Body Motion. In *2013 Humaine Association Conference on Affective Computing and Intelligent Interaction*. 349–355. <https://doi.org/10.1109/ACII.2013.64>
- [17] Harry J. Griffin, Min. S. Hane Aung, Bernadino Romera-Paredes, Ciaran McLoughlin, Gary McKeown, William Curran, and Nadia Bianchi-Berthouze. 2015. Perception and Automatic Recognition of Laughter from Whole-Body Motion: Continuous and Categorical Perspectives. *IEEE Transactions on Affective Computing* 6, 2 (2015), 165–178. <https://doi.org/10.1109/TAFFC.2015.2390627>
- [18] Jiajing Guo and Susan R. Fussell. 2020. A Preliminary Study of Emotional Contagion in Live Streaming. In *Companion Publication of the 2020 Conference on Computer Supported Cooperative Work and Social Computing (Virtual Event, USA) (CSCW '20 Companion)*. Association for Computing Machinery, New York, NY, USA, 263268. <https://doi.org/10.1145/3406865.3418309>
- [19] Jeffrey T. Hancock, Kailyn Gee, Kevin Ciacio, and Jennifer Mae-Hwah Lin. 2008. I'm sad you're sad: emotional contagion in CMC. In *Proceedings of the 2008 ACM Conference on Computer Supported Cooperative Work (San Diego, CA, USA) (CSCW '08)*. Association for Computing Machinery, New York, NY, USA, 295298. <https://doi.org/10.1145/1460563.1460611>
- [20] Julian Hanich. 2014. Watching a film with others: towards a theory of collective spectatorship. *Screen* 55, 3 (09 2014), 338–359. <https://doi.org/10.1093/screen/hju026>
- [21] Michel-Pierre Jansen, Khiet P. Truong, and Dirk K. J. Heylen. 2021. How Familiarity Influences the Frequency, Temporal Dynamics and Acoustics of Laughter. In *2021 9th International Conference on Affective Computing and Intelligent Interaction (ACII)*. 1–8. <https://doi.org/10.1109/ACII52823.2021.9597414>
- [22] Antonios Kalloniatis and Panagiotis Adamidis. 2024. Computational humor recognition: a systematic literature review. *Artificial Intelligence Review* 58, 2 (20 Dec 2024), 43. <https://doi.org/10.1007/s10462-024-11043-3>
- [23] Joon Soo Lim, YoungChan Hwang, Seyun Kim, and Frank A. Biocca. 2015. How social media engagement leads to sports channel loyalty: Mediating roles of social presence and channel commitment. *Computers in Human Behavior* 46 (2015), 158–167. <https://doi.org/10.1016/j.chb.2015.01.013>
- [24] Zhi-Song Liu, Robin Courant, and Vicky Kalogeiton. 2023. FunnyNet: Audiovisual Learning of Funny Moments in Videos. In *Computer Vision – ACCV 2022*, Lei Wang, Juergen Gall, Tat-Jun Chin, Imari Sato, and Rama Chellappa (Eds.). Springer Nature Switzerland, Cham, 433–450. https://doi.org/10.1007/978-3-031-26316-3_26
- [25] Maurizio Mancini, Giovanna Varni, Donald Glowinski, and Gualtiero Volpe. 2012. Computing and Evaluating the Body Laughter Index. In *Human Behavior Understanding*, Albert Ali Salah, Javier Ruiz-del Solar, Çetin Meriçli, and Pierre-Yves Oudeyer (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 90–98. https://doi.org/10.1007/978-3-642-34014-7_8
- [26] Mark McGill, John H. Williamson, and Stephen Brewster. 2016. Examining The Role of Smart TVs and VR HMDs in Synchronous At-a-Distance Media Consumption. *ACM Trans. Comput.-Hum. Interact.* 23, 5, Article 33 (Nov. 2016), 57 pages. <https://doi.org/10.1145/2983530>
- [27] Radosław Niewiadomski, Maurizio Mancini, Giovanna Varni, Gualtiero Volpe, and Antonio Camurri. 2016. Automated Laughter Detection From Full-Body Movements. *IEEE Transactions on Human-Machine Systems* 46, 1 (2016), 113–123. <https://doi.org/10.1109/THMS.2015.2480843>
- [28] Radosław Niewiadomski and Catherine Pelachaud. 2012. Towards Multimodal Expression of Laughter. In *Intelligent Virtual Agents*, Yukiko Nakano, Michael Neff, Ana Paiva, and Marilyn Walker (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 231–244. https://doi.org/10.1007/978-3-642-33197-8_24
- [29] Badri N. Patro, Mayank Lunayach, Deepankar Srivastava, Sarvesh Sarvesh, Hunar Singh, and Vinay P. Nambodiri. 2021. Multimodal Humor Dataset: Predicting Laughter tracks for Sitcoms. In *2021 IEEE Winter Conference on Applications of Computer Vision (WACV)*. 576–585. <https://doi.org/10.1109/WACV48630.2021.00062>
- [30] Tekla S. Perry. 2016. Virtual reality goes social. *IEEE Spectrum* 53, 1 (January 2016), 56–57. <https://doi.org/10.1109/MSPEC.2016.7367470>
- [31] Robert R. Provine. 1992. Contagious laughter: Laughter is a sufficient stimulus for laughs and smiles. *Bulletin of the Psychonomic Society* 30, 1 (01 Jul 1992), 1–4. <https://doi.org/10.3758/BF03330380>
- [32] Kevin S. Berbaum Robert S. Kennedy, Norman E. Lane and Michael G. Lilienthal. 1993. Simulator Sickness Questionnaire: An Enhanced Method for Quantifying Simulator Sickness. *The International Journal of Aviation Psychology* 3, 3 (1993), 203–220. https://doi.org/10.1207/s15327108ijap0303_3
- [33] Sylvia Rothe, Alexander Schmidt, Mario Montagud, Daniel Buschek, and Heinrich Hußmann. 2021. Social viewing in cinematic virtual reality: a design space for social movie applications. *Virtual Reality* 25, 3 (01 Sep 2021), 613–630. <https://doi.org/10.1007/s10055-020-00472-4>
- [34] Willibald Ruch and Paul Ekman. 2001. *THE EXPRESSIVE PATTERN OF LAUGHTER*. World Scientific, 426–443. https://doi.org/10.1142/9789812810687_0033
- [35] Willibald Friedrich Ruch, Jennifer Hofmann, and Tracey Platt. 2013. Investigating facial features of four types of laughter in historic illustrations. *The European Journal of Humour Research* 1, 1 (Mar. 2013), 99118. <https://doi.org/10.7592/EJHR2013.1.1.ruch>
- [36] Willibald F Ruch, Tracey Platt, Jennifer Hofmann, Radosław Niewiadomski, Jérôme Urbain, Maurizio Mancini, and Stéphane Dupont. 2014. Gelotophobia and the challenges of implementing laughter into virtual agents interactions. *Front. Hum. Neurosci.* 8 (Nov. 2014), 928. <https://doi.org/10.3389/fnhum.2014.00928>
- [37] Moria Smoski and Jo-Anne Bachorowski. 2003. Antiphonal laughter between friends and strangers. *Cognition and Emotion* 17, 2 (2003), 327–340. <https://doi.org/10.1080/026999303002296>
- [38] Keita Suzuki, Masanori Yokoyama, Shigeo Yoshida, Takayoshi Mochizuki, Tomohiro Yamada, Takuji Narumi, Tomohiro Tanikawa, and Michitaka Hirose. 2017. FaceShare: Mirroring with Pseudo-Smile Enriches Video Chat Communications. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (Denver, Colorado, USA) (CHI '17)*. Association for Computing Machinery, New York, NY, USA, 53135317. <https://doi.org/10.1145/3025453.3025574>
- [39] Kata Szita, Wyatt Moss-Wellington, Xiaolin Sun, and Eugene Ch'ng. 2024. Going to the movies in VR: Virtual reality cinemas as alternatives to in-person co-viewing. *Int. J. Hum.-Comput. Stud.* 181, C (Jan. 2024), 11 pages. <https://doi.org/10.1016/j.ijhcs.2023.103150>
- [40] Enrico Terrone. 2021. The Audience Effect. On the Collective Cinema Experience. *The British Journal of Aesthetics* 62, 1 (07 2021), 151–154. <https://doi.org/10.1093/aesthj/ayab013>
- [41] Jacob O. Wobbrock, Leah Findlater, Darren Gergle, and James J. Higgins. 2011. The aligned rank transform for nonparametric factorial analyses using only anova procedures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (Vancouver, BC, Canada) (CHI '11)*. Association for Computing Machinery, New York, NY, USA, 143146. <https://doi.org/10.1145/1978942.1978963>
- [42] Detai Xin, Shinnosuke Takamichi, Ai Morimatsu, and Hiroshi Saruwatari. 2023. Laughter Synthesis using Pseudo Phonetic Tokens with a Large-scale In-the-wild Laughter Corpus. In *Proc. Interspeech*. <https://doi.org/10.21437/interspeech.2023-806>
- [43] Hiromu Yakura and Masataka Goto. 2020. Enhancing Participation Experience in VR Live Concerts by Improving Motions of Virtual Audience Avatars. In *2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 555–565. <https://doi.org/10.1109/ISMAR50242.2020.00083>
- [44] Shigeo Yoshida, Tomohiro Tanikawa, Sho Sakurai, Michitaka Hirose, and Takuji Narumi. 2013. Manipulation of an emotional experience by real-time deformed facial feedback. In *Proceedings of the 4th Augmented Human International Conference (Stuttgart, Germany) (AH '13)*. Association for Computing Machinery, New York, NY, USA, 3542. <https://doi.org/10.1145/2459236.2459243>
- [45] Richard David Young and Margaret Frye. 1966. Some are Laughing; Some are Not—Why? *Psychological Reports* 18, 3 (1966), 747–754. <https://doi.org/10.2466/pr0.1966.18.3.747>